

OpenEd Design System and Experience Principles

Purpose

The current FME prototype proves useful structure but has visual issues: the dark theme feels heavy, the UI is flat, lesson screens can feel repetitive, and the product lacks enough visual excitement. OpenEd v1 should establish a broader, warmer, more flexible design language for serious learning.

Design strategy

OpenEd should feel like:

- A trusted learning studio.
- A guided workspace.
- A calm AI-native product.
- A platform for serious but accessible learning.

It should not feel like:

- A government dashboard.
- A flat LMS.
- A dark cybersecurity product.
- A generic SaaS template.
- A course PDF viewer.

Theme direction

Light-first

Light mode should be the default for learning. It improves readability, reduces fatigue, and feels less intimidating.

Suggested tokens:

Token	Value	Use
Canvas	#F8FAFC	App background

Token	Value	Use
Surface	#FFFFFF	Cards and panels
Soft surface	#F1F5F9	Secondary panels
Primary text	#0F172A	Main text
Secondary text	#475569	Supporting text
Border	#E2E8F0	Subtle dividers
Primary	#2563EB	Actions and progress
Evidence teal	#0F766E	Sources/proof
Tutor violet	#6D28D9	AI tutor and explanations
Warm accent	#F97316	Human/creative moments

Dark mode rebuild

The current dark mode should be softened. Avoid pure black, harsh saturated colors, and too many boxed panels.

Suggested tokens:

Token	Value	Use
Canvas	#0B1120	App background
Surface	#111827	Main cards
Soft surface	#1E293B	Secondary panels
Elevated	#273449	Drawers/modals
Text	#F8FAFC	Primary text
Muted text	#CBD5E1	Secondary text
Border	#334155	Dividers
Primary	#60A5FA	Actions
Evidence	#2DD4BF	Sources/proof
Tutor	#A78BFA	AI tutor

Visual excitement without clutter

Learners like moments of visual interest. OpenEd should add visual excitement through learning-native elements, not decorative noise.

Use:

- Progress maps.
- Animated but subtle learning milestones.
- Concept diagrams.
- Before/after examples.
- Visual tutor sketches.
- SkillProof cards.
- Mini simulations.
- Course identity covers.
- Achievement moments.
- Warm illustration spots.

Avoid:

- Random gradients.
- AI brain/robot imagery.
- Overcrowded dashboards.
- Neon accents.
- Excessive shadows.
- Decorative charts with no meaning.

Typography

Use a simple family stack:

- Inter or similar for UI and body.
- IBM Plex Mono for code, metadata, IDs, timestamps.
- Optional serif only for editorial quotes or educator notes.

Reading rules:

- Body: 16px minimum desktop/tablet, 15-16px mobile.
- Line height: 1.55-1.7 for reading.
- Lesson text column: 680-760px max on desktop.
- Avoid dense all-caps metadata.

Layout rules

Desktop

- Top bar: 64px.
- Left rail: 72px collapsed, 280px expanded.
- Main lesson: fluid center.
- Tutor panel: 340-380px.
- Use 24px and 32px spacing.

Tablet

- Top bar remains.
- Left rail collapses to drawer.
- Tutor can be side-by-side or drawer depending width.

Mobile

- Bottom nav or compact top nav.
- Tutor as bottom sheet.
- Sources as full-screen drawer.
- Artifact builder uses step-by-step flow.
- Avoid desktop tables; use stacked cards.

Component system

Core components:

- CourseCard.
- LessonCard.
- LessonWorkspace.
- VideoReferencePlayer.
- TranscriptDrawer.
- SourceCard.
- AITutorPanel.
- TutorPromptChip.
- PracticeCard.
- QuizCard.
- FeedbackPanel.

- ArtifactBuilder.
- RubricDrawer.
- SkillProofCard.
- PublishReadinessPanel.
- ContentQACard.

Lesson template variation

Every lesson should not look identical. Use template variants:

Lesson type	Primary visual
Concept lesson	Diagram + tutor explanation
Case lesson	Storyboard + decision point
Tool lesson	Screencast/reference media + checklist
Practice lesson	Interactive task + feedback
Project lesson	Artifact builder + rubric
Review lesson	Knowledge map + quiz

Design acceptance criteria

- Learner can identify primary action in 3 seconds.
- Dark mode no longer feels visually heavy.
- Lesson screens do not all look like the same card stack.
- The AI tutor feels integrated, not bolted on.
- Course sources feel trustworthy and easy to inspect.
- The product works on phone, tablet, and desktop without hidden core functionality.

Source notes

This document is grounded in the current OpenEd/FME project context and the following reference set where applicable:

- Uploaded Frontier Model Evaluations prototype code bundle, latest inspected bundle: Design System for Frontier Model Evaluation (5).zip.
- Uploaded Frontier Model Evaluations 51-hour curriculum and Module 1 foundations files.

- Coursera, Udemy, DataCamp, Moodle, Khanmigo, Duolingo public documentation and product pages.
- Supabase Auth and Row Level Security documentation.
- OpenAI API key safety guidance, MDN Web APIs, and YouTube IFrame Player API documentation.
- Bloom's 2 sigma problem, intelligent tutoring systems research, CourseAssist, LearnLM-Tutor, DCCI, and UNESCO guidance for generative AI in education.

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